

ORIGINAL CONTRIBUTION

Role of Perfusion Parameters on Outcomes and Safety of Endovascular Therapy in Posterior Cerebral Artery Stroke

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BACKGROUND: The optimal management of isolated posterior cerebral artery occlusion (iPCAO) remains unclear. We investigated whether baseline perfusion imaging parameters are associated with clinical outcomes and whether they modify the association between endovascular therapy (EVT) and outcomes in iPCAO.

METHODS: This prespecified secondary analysis of the international, multicenter, observational PLATO (Posterior Cerebral Artery Occlusion) registry (35 centers, 10 countries, 2015–2025) included consecutive adults with unilateral iPCAO and baseline perfusion imaging (computed tomography or magnetic resonance imaging) with reconstructed parameters. Primary end point was an excellent 90-day outcome (modified Rankin Scale score, 0–1). Perfusion parameters included hypoperfusion volume, infarct core volume, and mismatch ratio. The primary analysis used multivariable mixed-effects regression models (center as random effect) to assess associations between perfusion parameters and outcomes, adjusting for age, sex, treatment year, prestroke modified Rankin Scale score, baseline National Institutes of Health Stroke Scale score, diabetes, stroke cause, posterior circulation Acute Stroke Prognosis Early Computed Tomography Score, occlusion site, intravenous thrombolysis, and onset-to-door time. To test whether the association between EVT and outcomes varies according to baseline perfusion parameters, we evaluated treatment-by-perfusion interactions by including interaction terms

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(treatment×perfusion parameter) in inverse probability of treatment weighting–adjusted models, with results expressed as ratios of odds ratios (ORs).

RESULTS: Of 1811 patients with iPCAO, 443 met inclusion criteria (median age, 74 years; 41.8% female). Larger hypoperfusion volume was associated with lower odds of excellent outcome (adjusted OR, 0.72 [95% CI, 0.58–0.89] per 1-unit increase in natural logarithm–transformed volume). No interaction between perfusion parameters and EVT was observed for the primary outcome. However, increasing core volume was associated with a progressively less favorable modified Rankin Scale score shift (ratio of OR, 0.66 [95% CI, 0.48–0.90]; $P_{\text{interaction}}=0.009$) and higher mortality (ratio of OR, 1.82 [95% CI, 1.10–3.03]; $P_{\text{interaction}}=0.021$) with EVT compared with medical management. Increasing hypoperfusion volume was associated with a higher risk of symptomatic intracranial hemorrhage with EVT (ratio of OR, 10.15 [95% CI, 1.06–96.93]; $P_{\text{interaction}}=0.044$).

CONCLUSIONS: In iPCAO, perfusion imaging provides independent prognostic information but does not identify patients with potential benefit from EVT and may instead indicate those at higher procedural risk.

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GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: basilar artery ■ endovascular procedures ■ ischemic stroke ■ perfusion imaging ■ retrospective studies

Nonstandard Abbreviations and Acronyms

CT	computed tomography
DMVO	distal and medium vessel occlusion
EVT	endovascular therapy
iPCAO	isolated posterior cerebral artery occlusion
IPTW	inverse probability of treatment weighting
IQR	interquartile range
LVO	large vessel occlusion
MM	medical management
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
PCA	posterior cerebral artery
PLATO	Posterior Cerebral Artery Occlusion
ROR	ratio of odds ratios
siCH	symptomatic intracerebral hemorrhage

Isolated posterior cerebral artery occlusions (iPCAOs) account for up to 7% of all arterial occlusive ischemic strokes.¹ Given both their frequency and the substantial impact on quality of life, especially in patients with visual field impairment,² there is an urgent need for well-defined and effective treatment strategies.

Endovascular therapy (EVT) is an established standard of care for large vessel occlusions (LVOs) in the anterior circulation and in selected patients with basilar artery occlusion with moderate-to-severe deficits.^{3–8} Recent distal and medium vessel occlusion (DMVO) trials failed to demonstrate a benefit of EVT compared with standard medical management (MM),^{9,10} whereas one trial showed benefit of EVT in patients with higher baseline severity

deficit.¹¹ Patients with iPCAO were excluded from pivotal LVO trials and underrepresented in subsequent DMVO trials, and a definitive conclusion regarding the efficacy of EVT in this group of patients is lacking. Retrospective studies have not demonstrated a consistent benefit of EVT in iPCAO.^{12–15} However, a prior subanalysis of the PLATO study (Posterior Cerebral Artery Occlusion),¹⁶ the largest patient-level study on iPCAO to date, suggested that patients presenting with moderate-to-severe neurological deficits may benefit from EVT.¹⁷

Beyond baseline stroke severity, there is a lack of reliable predictors to identify patients who are most likely to benefit from EVT. Perfusion imaging metrics could be promising in this regard.¹⁸ Distinguishing between potentially salvageable tissue and established infarction with perfusion imaging can help to identify patients who are likely to benefit from EVT.¹⁹ In patients with suspected acute ischemic stroke and LVO presenting within 6 to 24 hours of last known well, current European Stroke Organisation guidelines recommend perfusion imaging selection to guide reperfusion therapy, and American Heart Association/American Stroke Association guidelines suggest it as a potentially useful tool.^{7,20} The use of perfusion imaging in DMVOs is less well-established, but previous observations suggested its potential advantage in identifying a subgroup of patients with DMVOs who could benefit from EVT despite the neutral results of recent trials.²¹ No large study has investigated the use of perfusion imaging in patients with iPCAOs, and data for advanced imaging in this population remain scarce.^{18,22}

In this prespecified secondary analysis of the PLATO study, we aim to investigate whether baseline computed tomography (CT)–based or magnetic resonance imaging–based perfusion imaging characteristics (core volume, hypoperfusion volume, and mismatch ratio) are associated with 3-month outcomes, primarily excellent

functional outcome (modified Rankin Scale [mRS] score, 0–1, or return to baseline), and if they modify the association between EVT and these outcomes in patients with iPCAO. We hypothesized that patients with iPCAO with favorable perfusion profiles (smaller ischemic core volumes and higher mismatch ratios) would present better clinical outcomes and greater benefit of EVT compared with those with less favorable profiles.

METHODS

Ethics

The PLATO study (Unique identifier: NCT05291637) was approved by the ethics committees or institutional review boards of all the centers involved. As the study was retrospective and the data were anonymized, no written informed consent was collected. The data were derived from the PLATO registry and stored in a password-protected database. Our reporting follows the Strengthening the Reporting of Observational Studies in Epidemiology guidelines ([Supplemental Material](#)). Anonymized data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Design and Population

This is a secondary analysis of the international, multicenter cohort PLATO study, which comprises consecutive adult patients (aged ≥ 18 years) with iPCAO. The analysis was conducted on the expanded PLATO registry and included consecutive patients treated between January 2015 and June 2025. Since the initial analysis,¹⁶ the PLATO study expanded with an additional 8 sites, totaling 35 sites across 10 countries in Europe, North America, and Asia. For this analysis, we included only patients from centers that routinely performed perfusion imaging, defined as availability of perfusion imaging in at least one-third of patients with iPCAO, and could provide reconstructed perfusion parameters. Among patients treated at these centers, we applied the following inclusion criteria: (1) acute ischemic stroke due to unilateral iPCAO (P1–P3 or fetal posterior cerebral artery [PCA]), (2) admission within 24 hours of symptom onset, (3) a prestroke mRS score of 0 to 3, and (4) baseline perfusion imaging (CT perfusion or magnetic resonance perfusion-weighted imaging) with available reconstructed ischemic core and hypoperfusion volumes. The exclusion criteria were (1) bilateral PCA occlusion, any concomitant basilar artery occlusion, or other vessel occlusion outside the PCA territory and (2) missing covariate data required for the adjusted analyses. Patients received either EVT or MM, with or without intravenous thrombolysis, based on the treating physicians' individual decisions.

Data Collection

Data were collected at each participating center and compiled in the updated PLATO registry. In addition to previously reported demographics, imaging variables, and clinical and safety outcomes, the current data set included perfusion imaging parameters, derived from CT perfusion or magnetic resonance perfusion-weighted imaging. Perfusion data were processed locally at the participating centers using automated

or semiautomated software, such as RAPID, VizAI, Brainomix, and Philips Healthcare Brilliance Workspace Portal, and the software used for each patient was recorded. Perfusion parameters of interest included (1) ischemic core volume (mL), defined as a relative cerebral blood flow $<30\%$ or an apparent diffusion coefficient value $<620 \times 10^{-6} \text{ mm}^2/\text{s}$; (2) hypoperfusion volume (mL), defined as time-to-maximum of the residue function $>6 \text{ s}$; (3) the mismatch ratio, calculated as the ratio of hypoperfusion volume to ischemic core volume; and (4) mismatch volume (mL), defined as the difference between hypoperfusion volume and ischemic core volume.

Outcome Measures

The 90-day mRS score was assessed at each participating center by trained and certified personnel according to local clinical practice, either during in-person visits or structured telephone interviews.

The primary end point was an excellent outcome at 90 days, defined as an mRS score of 0 to 1 or return to baseline mRS score in patients with a prestroke mRS score >1 . Secondary end points included (1) functional outcome at 90 days, defined as an ordinal shift in mRS score; (2) functional independence at 90 days, defined as mRS score 0 to 2 or return to baseline mRS score in patients with prestroke mRS score >2 ; (3) early neurological improvement at 24 hours or discharge, defined as ≥ 2 -point decrease in National Institutes of Health Stroke Scale (NIHSS) score; (4) symptomatic intracranial hemorrhage (sICH), defined as any parenchymal hemorrhage type 2, subarachnoid hemorrhage, or intraventricular hemorrhage, and judged to be causative of a ≥ 4 -point NIHSS score deterioration or death²³; (5) 90-day mortality; and (6) 90-day visual recovery.

Statistical Analysis

Descriptive statistics were used to compare baseline characteristics between included and excluded patients and between treatment groups (EVT versus MM). Continuous variables are presented as medians with interquartile ranges (IQRs) and were compared using the Wilcoxon rank-sum test. Categorical variables are presented as counts and percentages and were compared using the χ^2 or Fisher exact test, as appropriate. The primary outcome and binary secondary outcomes were analyzed using mixed-effect binary logistic regression, while the ordinal 90-day mRS score outcome was analyzed using mixed-effect ordinal logistic regression. For ordinal logistic regression models, the proportional odds assumption was assessed, with no major violations observed. All models included a random intercept for study center. In all analyses, each perfusion parameter (hypoperfusion volume, core volume, and mismatch ratio) was analyzed in separate models to avoid collinearity between related imaging parameters. Perfusion variables were analyzed as continuous measures. The distribution of perfusion parameters was visually assessed, and patients with extreme hypoperfusion volumes $>140 \text{ mL}$ were excluded from perfusion analyses, as this threshold exceeds the expected volume of the PCA vascular territory reported in prior studies (maximum infarct volume, 128 mL).²⁴ After assessing the perfusion parameters distribution and linearity assumption of the regression models, natural logarithmic transformation of the perfusion parameters was applied ([Figure S1](#)).

Associations between perfusion parameters and outcomes were tested using mixed-effects regression models, including crude models with the perfusion variable of interest as the only independent variable, and multivariable models adjusted for treatment group and prespecified baseline covariates, including age, sex, treatment year, prestroke mRS score, baseline NIHSS score, diabetes, stroke cause, posterior circulation Acute Stroke Prognosis Early CT Score, occlusion site, intravenous thrombolysis, and onset-to-door time. Covariates were selected a priori based on clinical relevance and previous literature, and no data-driven variable selection procedure was applied. Multicollinearity among covariates was assessed using variance inflation factors, with values <2 indicating no evidence of problematic collinearity. Results of this analysis are reported as odds ratios (ORs) per 1-unit increase in natural logarithm-transformed volume (mL). For outcomes with a limited number of events (sICH and mortality), sensitivity analyses were performed using a parsimonious adjustment set including age, prestroke mRS score, baseline NIHSS score, and intravenous thrombolysis to reduce the risk of overfitting.

The associations between treatment and outcomes were assessed using crude models and inverse probability of treatment weighting (IPTW)-adjusted models to address baseline imbalances between treatment groups. The propensity score was estimated using binary logistic regression, including prespecified covariates selected a priori based on clinical relevance and potential association with treatment allocation and outcomes. These included age, sex, occlusion site, prestroke mRS score, baseline NIHSS score, year of treatment, use of intravenous thrombolysis, posterior circulation Acute Stroke Prognosis Early CT Score, door-to-needle time, hypertension, stroke cause, atrial fibrillation, and diabetes. Covariate balance before and after weighting was assessed using standardized mean differences, with values <0.1 considered indicative of adequate balance. Overlap of propensity scores between treatment groups and the presence of extreme weights were also evaluated.

The potential modification of the association between treatment group and outcomes by perfusion parameters was evaluated by adding interaction terms (treatment $\times$ perfusion variables) to the IPTW-adjusted models. For graphical representation, we plotted the ORs and 95% CIs across the range of each perfusion parameter. Predicted outcome probabilities were derived from the same models across the observed range of each perfusion parameter, separately for EVT and MM, with CIs obtained from model-based standard errors.

As an exploratory analysis, to account for the absolute volume of salvageable tissue and complement the analyses on mismatch ratio, we evaluated the interaction between mismatch volume and EVT in IPTW-adjusted models further adjusted for core volume.

As a second exploratory analysis, we repeated the main interaction analyses between hypoperfusion volume, core volume, mismatch ratio, and treatment in patients with baseline NIHSS score >6 .

Further analyses included multivariable linear regression models examining independent associations between prespecified baseline characteristics and baseline perfusion parameters, and IPTW-adjusted regression models evaluating the association between successful EVT reperfusion defined

as modified Thrombolysis in Cerebral Infarction score $\geq 2b$, and clinical outcomes, with interaction terms testing effect modification by perfusion parameters.

No formal adjustment for multiple comparisons was applied, as this was an observational, hypothesis-generating study. Although a primary outcome and key hypothesis were prespecified, analyses of secondary outcomes and additional models were considered exploratory and interpreted with caution. Statistical significance was defined as a 2-sided $P < 0.05$, including for interaction terms. Analyses were conducted using R, version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria). The prespecified study protocol and statistical analysis plan were registered on the Open Science Framework before accessing the data set (<https://osf.io/62mwt>; registered on November 19, 2025).

RESULTS

Participants

Of 1811 patients with iPCAO included in the PLATO registry, 831 were treated at 9 centers that routinely performed perfusion imaging and provided reconstruction parameters.

Among them, 365 patients were excluded, including 108 not undergoing perfusion imaging and 114 undergoing perfusion imaging but lacking reconstruction data (detailed reasons for exclusion displayed in Figure 1). Of the 466 patients who met the study's eligibility criteria, 443 had complete covariate data for adjusted regression analyses or IPTW and were included in the final study population (Figure 1). Baseline characteristics of patients who underwent perfusion imaging (regardless of reconstruction data availability) and those who did not are displayed in Table S1. A comparison of patients with perfusion imaging with and without reconstructed perfusion data is displayed in Table S2.

Baseline Characteristics

The 443 included patients had a median age of 74 (IQR, 63–82) years, 185 (41.8%) were female, 267 (60.3%) received MM, and 176 underwent EVT (39.7%). Compared with patients receiving MM, those treated with EVT were more recently enrolled (later inclusion date), presented with a higher median baseline NIHSS score (7 versus 4), and more frequently had large-artery atherosclerosis cause (17% versus 9%) but less often a cardioembolic stroke (33% versus 43.1%; Table 1). The occlusion site was more frequently located in the P1 segment and less frequently in the P2 or P3 segments. Overall, the median core volume was 0 (IQR, 0–5) mL, hypoperfusion volume was 27 (IQR, 12–45.7) mL, mismatch volume was 22 (IQR, 9–40.4) mL, and the mismatch ratio was 12 (IQR, 3–32). Patients undergoing EVT had a higher hypoperfusion volume (median, 31 versus 22 mL;

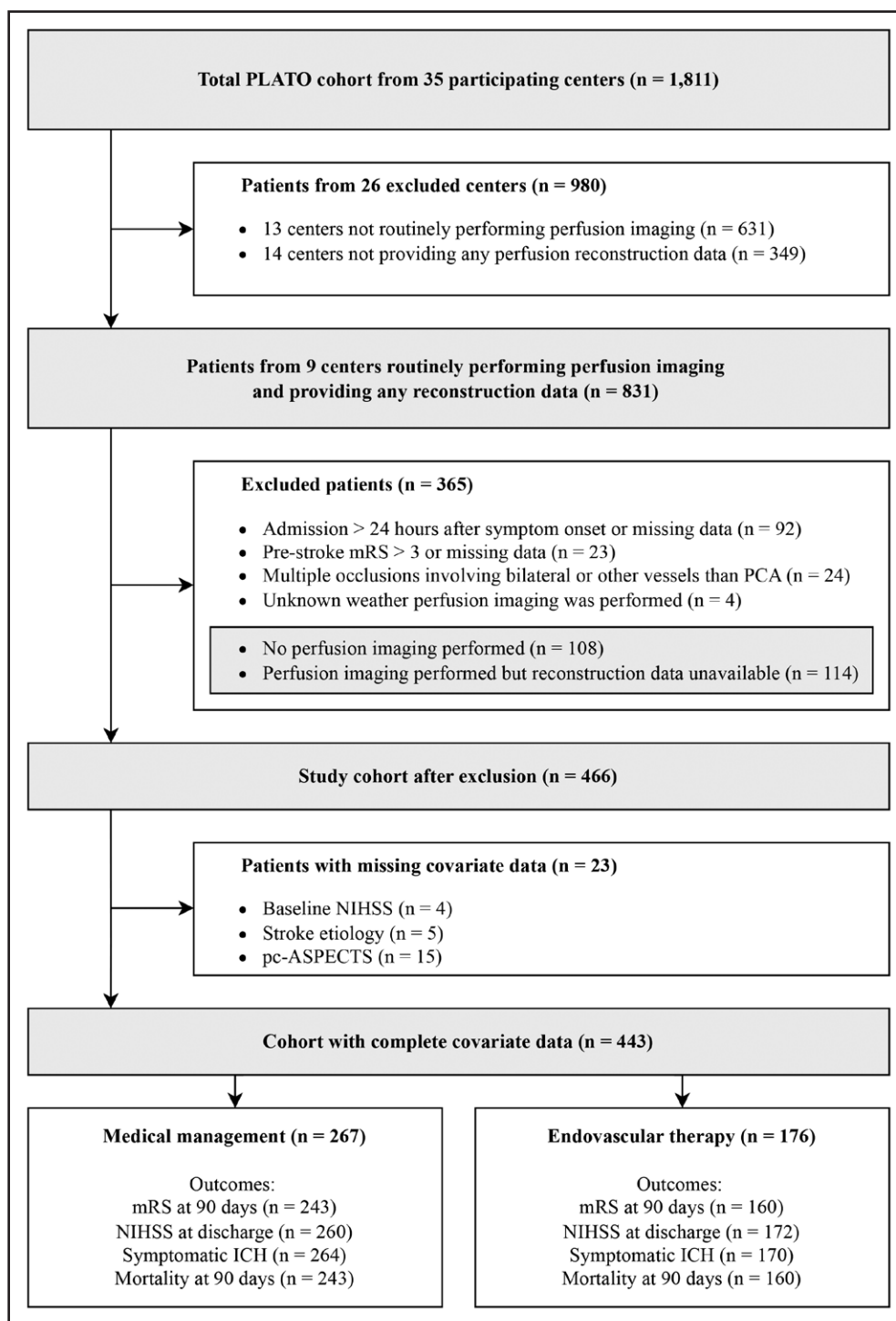


Figure 1. Study flowchart with patient selection process.

ICH indicates intracerebral hemorrhage; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; PCA, posterior cerebral artery; pc-ASPECTS, posterior circulation Acute Stroke Prognosis Early Computed Tomography Score; and PLATO, Posterior Cerebral Artery Occlusion.

$P=0.001$), larger mismatch volume (median, 27 versus 18 mL; $P<0.001$), and higher mismatch ratio (median, 18.6 versus 8.0; $P<0.001$), while core volume was similar in the 2 groups ($P=0.073$; Table 1). The proportion of missing values in baseline characteristics is shown in Table S3 and Figure S2.

Outcome Comparison Between Treatment Groups

The 90-day mRS score was available in 403 of 443 patients (91%). Data required for secondary outcomes (NIHSS score at discharge, sICH, and 90-day mortality)

Table 1. Baseline Characteristics of Patients With Isolated Posterior Cerebral Artery Occlusion Treated With MM Versus EVT

	Overall (n=443)	MM (n=267)	EVT (n=176)	P value
Demographics and clinical characteristics				
Age, y; median (IQR)	74 (63–82)	75 (64–82)	73 (62–81.6)	0.328
Female sex, n (%)	185/443 (41.8)	114/267 (42.7)	71/176 (40.3)	0.694
Median year of treatment, 2000s (IQR)	2020 (2018–2022)	2020 (2017–2022)	2020 (2019–2022)	0.001
Admission directly to EVT center, n (%)	353/396 (89.1)	202/223 (90.6)	151/173 (87.3)	0.377
Median prestroke mRS score (IQR)	0 (0–2)	0 (0–1)	0 (0–2)	0.872
Median baseline NIHSS score (IQR)	5 (3–9)	4 (2–8)	7 (4–10)	<0.001
Baseline visual field deficit, n (%)	235/407 (57.7)	147/260 (56.5)	88/147 (59.9)	0.584
Median admission SBP, mm Hg (IQR)	154 (134–173)	155 (136–176)	153.5 (132.9–171.1)	0.364
Stroke cause, n (%)				
Large-artery atherosclerosis	54/443 (12.2)	24/267 (9)	30/176 (17)	0.017
Cardioembolism	173/443 (39.1)	115/267 (43.1)	58/176 (33)	0.042
Other determined cause	57/443 (12.9)	39/267 (14.6)	18/176 (10.2)	0.229
Undetermined cause	159/443 (35.9)	89/267 (33.3)	70/176 (39.8)	0.20
Vascular risk factors and premedication, n (%)				
Hypertension	273/404 (67.6)	170/249 (68.3)	103/155 (66.4)	0.786
Diabetes	101/443 (22.8)	62/267 (23.2)	39/176 (22.2)	0.885
Hyperlipidemia	223/443 (50.3)	136/267 (50.9)	87/176 (49.4)	0.831
Current smoker	81/399 (20.3)	45/246 (18.3)	36/153 (23.5)	0.256
Previous stroke	71/439 (16.2)	45/264 (17)	26/175 (14.9)	0.633
Coronary heart disease	85/443 (19.2)	54/267 (20.2)	31/176 (17.6)	0.576
Peripheral artery disease	21/425 (4.9)	13/267 (4.9)	8/158 (5.1)	1.000
Atrial fibrillation	119/443 (26.9)	73/267 (27.3)	46/176 (26.1)	0.865
Dialysis	41/424 (9.7)	21/266 (7.9)	20/158 (12.7)	0.151
Oral anticoagulation	56/443 (12.6)	32/267 (12)	24/176 (13.6)	0.715
Statin	133/443 (30)	88/267 (33)	45/176 (25.6)	0.120
Initial imaging				
MRI, n (%)	73/443 (16.5)	37/267 (13.9)	36/176 (20.4)	0.089
CT, n (%)	380/443 (85.8)	235/267 (88)	145/176 (82.4)	0.128
Median pc-ASPECTS (IQR)	9 (9–10)	9 (9–10)	10 (9–10)	0.193
Hyperdense PCA on CT or blooming sign on MRI, n (%)	99/375 (26.4)	43/201 (21.4)	56/174 (32.2)	0.025
Occlusion site, n (%)				
P1	135/443 (30.5)	48/267 (18)	87/176 (49.4)	<0.001
P2	273/443 (61.6)	187/267 (70)	86/176 (48.9)	<0.001
P3	32/443 (7.2)	30/267 (11.2)	2/176 (1.1)	<0.001
Fetal PCA	3/443 (0.7)	2/267 (0.8)	1/176 (0.6)	1.000
Perfusion software, n (%)				
RapidAI	393/440 (89.3)	232/264 (87.9)	161/176 (91.5)	0.298
Vizai	1/440 (0.2)	0/264 (0)	1/176 (0.6)	0.838
Brainomix	5/440 (1.1)	0/264 (0)	5/176 (2.8)	0.022
PHILIPS	10/440 (2.3)	10/264 (3.8)	0/176 (0)	0.022
Other	31/440 (7)	22/264 (8.3)	9/176 (5.1)	0.270
Median core volume,* mL (IQR)	0 (0–5)	0 (0–6)	0 (0–3.6)	0.073
Median hypoperfusion (Tmax >6 s) volume, mL (IQR)	27 (12–45.7)	22 (8.2–41)	31 (17.4–52)	0.001
Median mismatch volume,† mL (IQR)	22 (9–40.8)	18 (5–36.7)	27 (15–48.6)	<0.001
Median mismatch ratio‡ (IQR)	12 (3–32)	8 (1.8–27)	18.6 (5–38)	<0.001
Time metrics and procedural characteristics				
Intravenous thrombolysis, n (%)	248/443 (56)	156/267 (58.4)	92/176 (52.3)	0.238

(Continued)

Table 1. Continued

	Overall (n=443)	MM (n=267)	EVT (n=176)	P value
Median time from symptom onset to door, min (IQR)	148 (73–370)	145 (75.2–369.7)	149 (70–375.3)	0.782
Median door-to-needle time, min (IQR)	35 (25–56)	36 (25–57.1)	31.5 (24.4–55)	0.815
Median door-to-groin time, min (IQR)			96.5 (75–121)	
General anesthesia used, n (%)			82/169 (48.5)	
Median number of passes (IQR)			1 (1–2)	
First pass EVT method, n (%)				
None§			12/120 (10)	
Stent retriever			7/120 (5.8)	
Stent retriever+contact aspiration			64/120 (53.3)	
Contact aspiration			37/120 (30.8)	
Procedural complications, n (%)				
Iatrogenic dissection			4/167 (2.4)	
Downstream embolization in the same territory			9/167 (5.4)	
Embolization in a new territory			1/167 (0.6)	
Arterial perforation			2/167 (1.2)	
Vasospasm			1/167 (0.6)	
Hematoma at access site			1/167 (0.6)	
Reocclusion			5/64 (7.8)	
Final mTICI score, n (%)				
0-2a			37/174 (21.3)	
2b			34/174 (19.5)	
2c			24/174 (13.8)	
3			79/174 (45.4)	

Continuous variables are reported as the median and IQR, and categorical variables are reported as the number of patients and column percentages. CT indicates computed tomography; EVT, endovascular therapy; IQR, interquartile range; MM, medical management; MRI, magnetic resonance imaging; mRS, modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; PCA, posterior cerebral artery; pc-ASPECTS, posterior circulation stroke prognosis early computed tomography score; SBP, systolic blood pressure; and Tmax, time to maximum of the residual function.

*Defined as a relative cerebral blood flow of <30% or the apparent diffusion coefficient value <620×10⁻⁶ mm²/s.

†Defined as the difference between hypoperfusion volume and core volume.

‡Defined as the ratio of hypoperfusion volume to core volume.

§No maneuver performed because of spontaneous recanalization or clot migration, use of local thrombolysis or angioplasty without mechanical thrombectomy, failed vascular access, procedure stopped due to complications, or the absence of a specifically reported reason.

were available for 432 (97.5%), 434 (98%), and 403 (91%) patients, respectively. Information on final infarct volume and visual impairment at follow-up was available for 263 (59.4%) and 221 (49.9%) patients, respectively.

No significant differences were observed in the crude rates of excellent outcome between the EVT and MM group (32.5% versus 38.3%; crude common OR, 0.77 [95% CI, 0.51–1.18]) nor after IPTW (common OR, 1.01 [95% CI, 0.64–1.57]; Table 2). After weighting, covariate balance between treatment groups improved substantially, with all standardized mean differences <0.1, and adequate overlap of propensity score distributions was observed (Figure S3). After IPTW adjustment, the associations of EVT with higher 90-day mRS scores, lower rates of functional independence, and more frequent early neurological improvement in the crude model were no longer significant (Table 2). However, EVT was associated with increased sICH (OR, 4.40 [95% CI, 1.01–19.11]) and 90-day mortality (OR, 1.93 [95% CI, 1.05–3.53]). In crude analyses, final infarct volume did

not differ between groups, whereas complete visual recovery was more frequent in the EVT group (Table 2).

Association Between Perfusion Parameters and Outcomes in the Overall Cohort

Associations between perfusion parameters and outcomes were assessed in the overall cohort, including both EVT and MM groups. From this analysis, 14 outlier patients with extreme hypoperfusion volumes >140 mL were excluded. ORs for perfusion parameters in this section are reported per 1-unit increase in natural logarithm-transformed volume (mL), which corresponds to a ≈2.7-fold increase in volume. Larger hypoperfusion volume was associated with lower odds of an excellent outcome both before and after adjustment for covariates (adjusted OR, 0.72 [95% CI, 0.58–0.89]). It was likewise associated with a less favorable mRS score shift (adjusted common OR, 0.72 [95% CI, 0.60–0.86]) and lower odds for functional independence (adjusted OR, 0.71 [95%

Table 2. Univariable Comparisons, Crude, and IPTW-Adjusted Associations Between Treatment Modality (EVT Versus MM) and Functional and Safety Outcomes

Outcome	MM (n=267)	EVT (n=176)	P value	Crude model, OR (95% CI)	IPTW model, OR (95% CI)
Primary outcome					
Excellent outcome at 90 d,* n (%)	93/243 (38.3)	52/160 (32.5)	0.282	0.77 (0.51–1.18)	1.01 (0.64–1.57)
Secondary outcomes					
mRS score at 90 d, median (IQR)	2 (1–3)	3 (1–4)	0.019	0.65 (0.45–0.93)†	0.76 (0.52–1.10)
Functional independence at 90 d,‡ n (%)	152/243 (62.5)	80/160 (50)	0.017	0.61 (0.40–0.92)†	0.71 (0.46–1.09)
Decrease in NIHSS score by ≥2 points,§ n (%)	112/260 (43.1)	103/172 (59.9)	0.001	1.95 (1.31–2.90)†	1.27 (0.84–1.91)
Symptomatic ICH, n (%)	4/264 (1.5)	7/170 (4.1)	0.170	3.19 (0.88–11.51)	4.40 (1.01–19.11)†
Mortality at 90 d, n (%)	24/243 (9.9)	30/160 (18.8)	0.016	2.11 (1.18–3.76)†	1.93 (1.05–3.53)†
Median final infarct volume, mL (IQR)	12 (2–30.8)	7.2 (1.4–26.8)	0.227		
Vision recovery at 90 d, n (%)					
Complete	49/221 (22.2)	39/114 (34.2)	0.025		
Partial	8/221 (3.6)	1/114 (0.9)	0.265		
Same	25/221 (11.3)	3/114 (2.6)	0.012		
Worse	35/221 (15.8)	14/114 (12.3)	0.478		
No visual deficit at onset	104/221 (47.1)	57/114 (50)	0.693		

Continuous variables are reported as the median and IQR, and categorical variables are reported as the number of patients and column percentages. IPTW was based on age, sex, occlusion site, prestroke mRS, baseline NIHSS score, year of treatment, intravenous thrombolysis, pc-ASPECTS, door-to-needle time, hypertension, stroke cause, atrial fibrillation, and diabetes. EVT indicates endovascular therapy; ICH, intracranial hemorrhage; IPTW, inverse probability of treatment weighting; IQR, interquartile range; MM, medical management; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; OR, odds ratio; and pc-ASPECTS, posterior circulation Acute Stroke Prognosis Early Computed Tomography Score.

*Defined as mRS score of 0–1 or return to prestroke mRS score at 90 d.

†Significant at $P < 0.05$ level.

‡Defined as mRS score of 0–2 or return to prestroke mRS score at 90 d.

§NIHSS score at discharge compared with NIHSS score at admission.

CI, 0.56–0.91]). Larger core volume was associated with lower odds of excellent outcome in unadjusted analysis (OR, 0.78 [95% CI, 0.65–0.94]) but not after multivariable adjustment. Larger core volume was associated with a less favorable mRS score shift both before and after adjustment (adjusted common OR, 0.80 [95% CI, 0.68–0.94]). No associations between perfusion parameters and sICH or mortality were observed, including in sensitivity analyses using parsimonious adjustment models (Table S4). No significant association was observed between mismatch ratio and any outcome in multivariable analysis (Table 3) across the full observed range of mismatch ratio values (0–138).

Interaction Analysis Between Perfusion Parameters and Treatment

No significant interaction between any perfusion parameter and EVT was observed for the primary outcome (Figure 2A and 2B; Figure S4). However, significant interactions between ischemic core volume and EVT were observed for secondary outcomes, as, at larger core volumes, EVT compared with MM was associated with progressively more unfavorable mRS score distributions (ratio of ORs [ROR], 0.66 [95% CI, 0.48–0.90]; $P_{\text{interaction}} = 0.009$; Figure 2C and 2D) and lower odds of functional independence (ROR, 0.66 [95% CI, 0.44–0.99]; $P_{\text{interaction}} = 0.043$;

Figure 2E and 2F; Figure S4). The association between EVT and sICH was modified by hypoperfusion volume (ROR, 10.15 [95% CI, 1.06–96.93]; $P_{\text{interaction}} = 0.044$), with a progressively higher risk of sICH associated with EVT at larger hypoperfusion volumes (Figure 3A and 3B). Similarly, the association between EVT and mortality was modified by core volume (ROR, 1.82 [95% CI, 1.10–3.03]; $P_{\text{interaction}} = 0.021$), with a higher mortality rate associated with EVT at larger core volumes (Figure 3E and 3F). No significant interactions were found for the mismatch ratio (Figure S5) or for mismatch volume in the exploratory analysis adjusted for core volume (Figure S6). The significances of all interaction terms from these analyses are summarized in Table S5 and Figures S7 and S8. In the exploratory analysis on the subgroup of patients with baseline NIHSS score >6 ($n=173$), a significant interaction was observed between hypoperfusion volume and EVT for mortality (ROR, 3.66 [95% CI, 1.09–12.29]; $P_{\text{interaction}} = 0.035$; Table S6), with larger volumes associated with increased odds of mortality in EVT-treated patients and a trend toward a less favorable mRS score shift (ROR, 0.59 [95% CI, 0.34–1.03]; $P_{\text{interaction}} = 0.061$).

Additional Analyses

Additional exploratory analyses are reported in the Supplemental Material, including associations between

Table 3. Crude and Multivariable Regression Analyses of the Association Between Perfusion Parameters and Functional and Safety Outcomes

Outcome	Hypoperfusion volume (ln)		Core volume (ln)		Mismatch ratio (ln)	
	Univariable model, OR (95% CI)	Multivariable model, aOR (95% CI)	Univariable model, OR (95% CI)	Multivariable model, aOR (95% CI)	Univariable model, OR (95% CI)	Multivariable model, aOR (95% CI)
Primary outcome						
Excellent outcome at 90 d*	0.68 (0.56–0.82)†	0.72 (0.58–0.89)†	0.78 (0.65–0.94)†	0.83 (0.67–1.03)	0.87 (0.75–1.02)	0.87 (0.72–1.04)
Secondary outcomes						
Favorable mRS score shift at 90 d	0.68 (0.58–0.81)†	0.72 (0.60–0.86)†	0.84 (0.73–0.97)†	0.80 (0.68–0.94)†	0.87 (0.76–0.99)†	0.91 (0.79–1.06)
Functional independence at 90 d‡	0.65 (0.52–0.80)†	0.71 (0.56–0.91)†	0.87 (0.73–1.03)	0.81 (0.65–1.01)	0.84 (0.72–0.98)†	0.90 (0.75–1.08)
Decrease in NIHSS score by ≥2 points§	1.06 (0.88–1.27)	0.83 (0.66–1.06)	0.88 (0.75–1.04)	0.87 (0.70–1.08)	1.11 (0.95–1.28)	0.95 (0.78–1.14)
Symptomatic ICH	1.51 (0.74–3.06)	1.31 (0.62–2.77)	1.26 (0.78–2.03)	1.65 (0.90–3.04)	1.06 (0.66–1.71)	0.90 (0.54–1.49)
Mortality at 90 d	1.39 (1.01–1.90)†	1.10 (0.77–1.57)	1.21 (0.96–1.53)	1.18 (0.89–1.58)	1.06 (0.85–1.33)	0.92 (0.71–1.20)

Adjusted models were controlled for age, sex, treatment year, prestroke mRS, baseline NIHSS score, diabetes, stroke cause, pc-ASPECTS, occlusion site, IVT, onset-to-door time, and EVT, with a random intercept for center. aOR indicates adjusted odds ratio; EVT, endovascular therapy; ICH, intracranial hemorrhage; IVT, intravenous thrombolysis; ln, natural logarithm; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; OR, odds ratio; and pc-ASPECTS, posterior circulation Acute Stroke Prognosis Early Computed Tomography Score.

*Defined as mRS score of 0–1 or return to prestroke mRS score at 90 d.

†Significant at $P < 0.05$ level.

‡Defined as mRS score of 0–2 or return to prestroke mRS score at 90 d.

§NIHSS score at discharge compared with NIHSS score at admission.

characteristics and perfusion parameters (Table S7), as well as the analyses of the interaction between perfusion parameters and successful reperfusion after EVT (Table S8; Figure S9).

DISCUSSION

This prespecified secondary analysis of the international, multicenter PLATO study showed that among patients presenting with iPCAO within 24 hours of symptom onset, perfusion parameters, mainly ischemic core volume, modified the association between EVT and clinical outcomes. In patients with larger ischemic core volumes, EVT was associated with similar rates of excellent outcome but worse overall 90-day disability, as well as higher risks of sICH and mortality compared with MM. In contrast, in patients with smaller core volumes, EVT was associated with similar mRS score distributions and comparable risks of mortality and sICH compared with MM. Importantly, across the spectrum of perfusion parameter values, we did not identify any range in which EVT was associated with more favorable outcomes compared with MM. Regardless of acute treatment, perfusion parameters demonstrated prognostic value, with smaller hypoperfusion and ischemic core volumes independently associated with better 90-day outcomes.

The findings of the present analysis extend previous observations from the PLATO collaboration. Compared with the primary PLATO analysis,¹⁶ the outcome comparison between treatment groups yielded overall similar results. The association between EVT and excellent

outcome was not observed in the present analysis, possibly reflecting the smaller sample size and different baseline imbalances between treatment groups. The similar mRS score distributions between EVT and MM, as well as the increased risks of sICH and mortality with EVT observed in both the original PLATO analysis and the present secondary analysis, are consistent with findings from recent randomized controlled trials in DMVOs.^{9,10} However, this body of evidence does not exclude the possibility that EVT may be safe and beneficial in carefully selected patients with iPCAO¹⁷ or DMVO in general.

By integrating baseline perfusion parameters, the present analysis adds an additional dimension to prior studies in the iPCAO population and represents, to our knowledge, the largest cohort of patients with iPCAO and available perfusion imaging to date. Contrary to our initial hypothesis, perfusion parameters did not modify the association between EVT and the primary outcome and were not proven to be useful for selecting patients with PCA occlusion who are likely to benefit from EVT. Instead, our results suggest that perfusion imaging may be better suited to identifying patients at increased risk of harm from EVT. In exploratory analyses, successful EVT-related recanalization was associated with less neurological improvement at larger ischemic core volumes, suggesting diminishing treatment effect with increasing core size. Together with the higher risks of sICH and mortality observed at larger core volumes, this may contribute to the less favorable mRS score distribution associated with EVT in this subgroup.



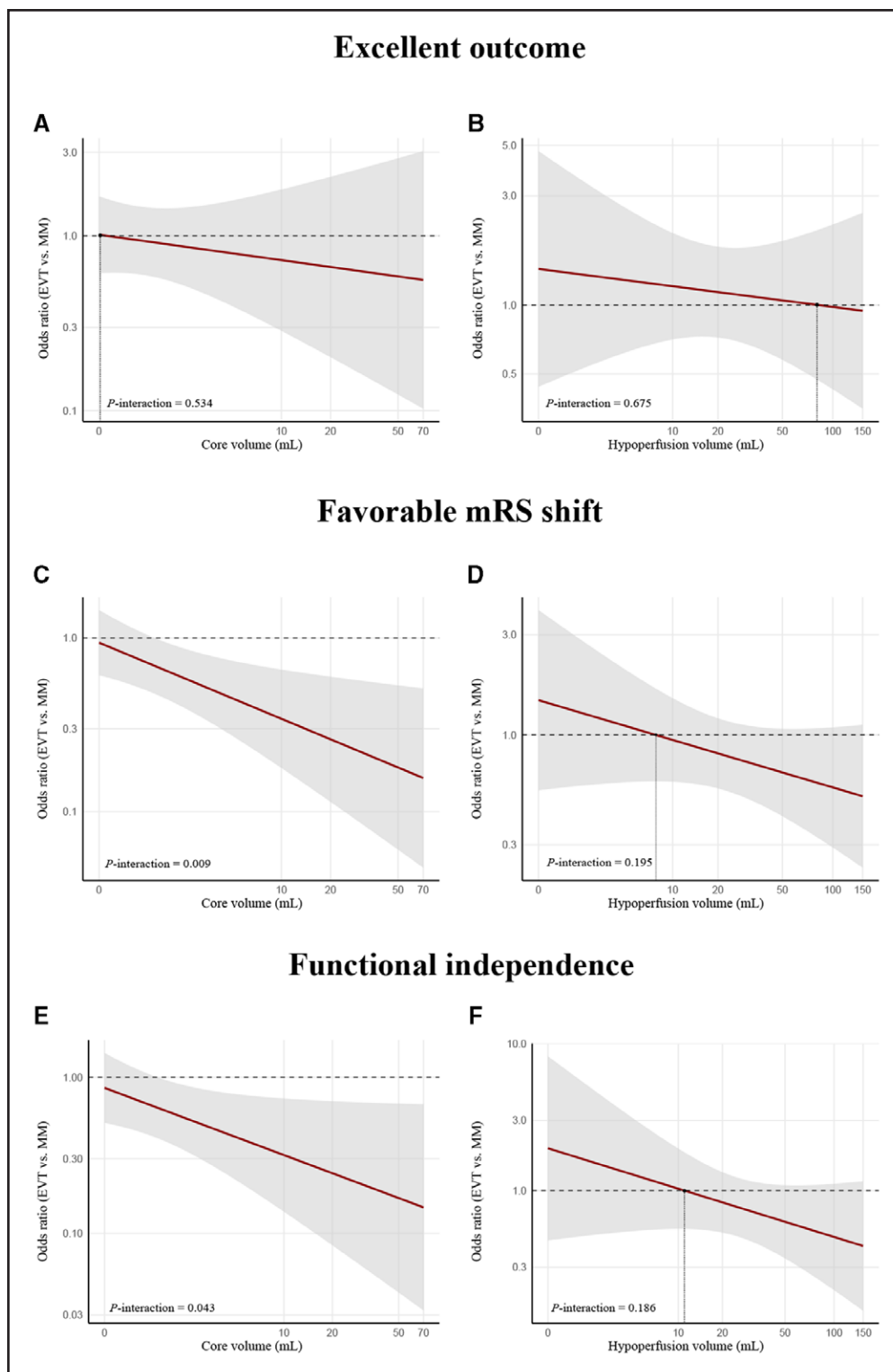


Figure 2. XXX.

Association between treatment group (endovascular therapy [EVT] vs medical management [MM]) and 3-month excellent outcome (**A** and **B**), modified Rankin Scale (mRS) score (**C** and **D**), and functional independence (**E** and **F**) according to continuous values of core and hypoperfusion volumes. Solid lines indicate estimated odds ratios (ORs) for EVT vs MM, and shaded areas represent 95% CIs. Estimates are derived from inverse probability of treatment weighting (IPTW)-adjusted regression models. The dashed horizontal line denotes an OR of 1. Values >1 indicate a higher likelihood of outcome with EVT, whereas values <1 indicate a higher likelihood with MM. Vertical dotted lines indicate the value of the perfusion parameter at which the estimated OR (EVT vs MM) crosses the line of no effect (OR, 1). The x axis is displayed on a logarithmic scale. P values refer to the interaction between treatment modality and the respective continuous perfusion parameter. IPTW was based on age, sex, occlusion site, prestroke mRS score, baseline National Institutes of Health Stroke Scale score, year of treatment, intravenous thrombolysis, posterior circulation Acute Stroke Prognosis Early Computed Tomography Score, door-to-needle time, hypertension, stroke cause, atrial fibrillation, and diabetes.

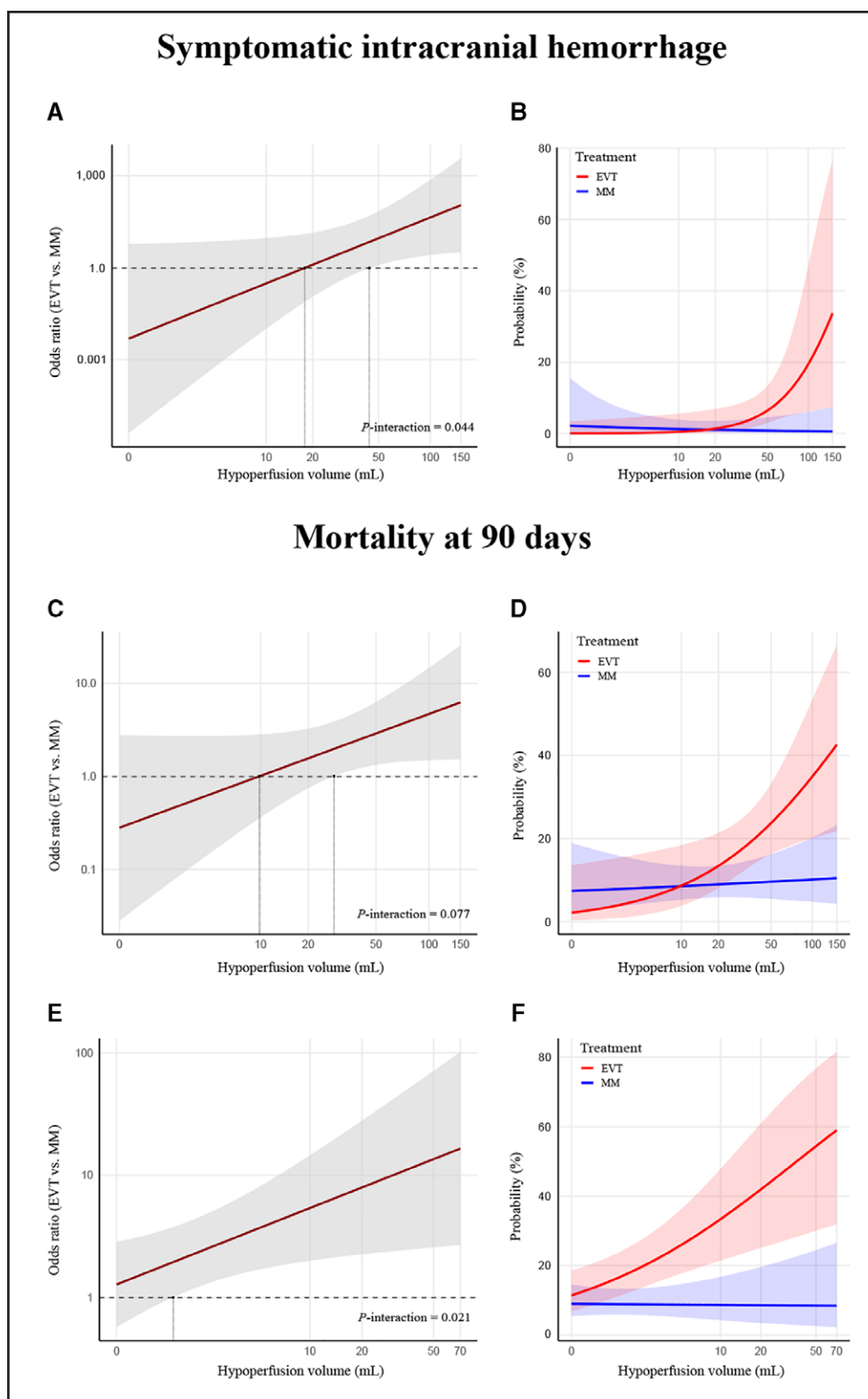


Figure 3. XXX.

Association between treatment group (endovascular therapy [EVT] vs medical management [MM]) and symptomatic intracerebral hemorrhage (**A** and **B**) and 3-month mortality (**C–F**). In **A**, **C**, and **E**, solid lines indicate estimated odds ratios (ORs) of EVT vs MM for the outcome across values of the perfusion parameter, and shaded areas indicate 95% CIs. Estimates are derived from inverse probability of treatment weighting–adjusted regression models. The dashed horizontal line denotes an OR of 1. Values >1 indicate a higher likelihood of outcome with EVT, whereas values <1 indicate a higher likelihood with MM. Vertical dotted lines indicate the value of the perfusion parameter at which the estimated OR (EVT vs MM) crosses the line of no effect (OR, 1). The x axis is displayed on a logarithmic scale. *P* values refer to the interaction between treatment modality and the respective continuous perfusion parameter. **B**, **D**, and **F**, Corresponding model-based predicted probabilities of each outcome for MM (blue) and EVT (red).

Consistent with our observation that, at larger hypoperfusion volumes, EVT is associated with a higher risk of sICH compared with MM, perfusion-derived indices have also been associated with increased risk of hemorrhagic transformation in anterior circulation DMVOs treated with EVT.²⁵ Other studies showed that in distal occlusion of the middle cerebral artery, larger hypoperfusion volumes are associated with a more beneficial effect of successful recanalization.¹⁸ At the same time, observations regarding the modifying role of perfusion parameters on EVT benefit in anterior circulation occlusions are heterogeneous. For example, mismatch volume has been shown to modify the effect of EVT in anterior circulation LVOs,²⁶ whereas no such modification was found in the pooled analysis of the HERMES or ATLAS collaborations.^{27,28} These analyses were based on randomized controlled trials with specific inclusion criteria, limiting direct comparability with our retrospective cohort, in addition to the different occlusion site.

Independent of treatment, perfusion parameters were significantly associated with several outcomes, in line with existing evidence. Larger ischemic core volumes have consistently been associated with worse functional outcomes in anterior circulation LVO stroke, whereas mismatch volume, similar to our findings on mismatch ratio, did not influence outcomes.^{27,29} Among patients with DMVOs, favorable perfusion profiles were likewise associated with better outcomes²¹ although all participants in this report underwent EVT, limiting comparison with our mixed cohort. In iPCAO, a small study suggested that larger hypoperfusion volumes involving midbrain-thalamic regions were independent predictors of poorer functional outcome after EVT,¹⁸ underscoring the importance of the regional distribution of affected brain tissue beyond the absolute volume metrics.

The distribution of perfusion parameters in our study was noteworthy, with a median ischemic core volume of 0 mL and substantially smaller hypoperfusion volumes than typically observed in patients with LVO. Therefore, target mismatch criteria derived from anterior circulation LVOs may not be applicable to iPCAOs. Although these findings are not unexpected, they underscore the need to define iPCAO-specific thresholds, as also highlighted in a recent analysis of anterior circulation DMVOs.³⁰

Despite limited supporting evidence, approximately half of the patients with iPCAO in the PLATO cohort underwent perfusion imaging, reflecting its perceived clinical relevance but also substantial intercenter variability in its use. While perfusion imaging may improve diagnostic accuracy to detect DMVOs, including PCAOs,^{31,32} no definitive evidence currently supports its use for EVT treatment selection or for predicting treatment benefit. Even in anterior circulation LVO, where perfusion imaging may play a role in individual treatment decisions,³³ evidence remains conflicting, and a large retrospective analysis suggests that CT perfusion-based selection for EVT

does not improve functional outcomes compared with noncontrast CT.³⁴ For iPCAO, there is currently no supporting evidence on whether or how imaging perfusion-based criteria should be applied to EVT decision-making.

Integrating the present findings with prior observations, including those from a previous PLATO analysis indicating that EVT may be beneficial primarily in patients with more severe baseline deficits,¹⁷ yields a potential clinical implication. While perfusion imaging does not identify patients likely to benefit from EVT, EVT may still be considered in selected patients with severe baseline deficits and relatively small ischemic core volumes, in whom the potential benefits may outweigh procedural risks.

Importantly, our analyses do not allow the identification of a clear perfusion-based cutoff separating the absence from the presence of harm associated with EVT, as the estimated point at which harm emerged varied across outcomes. For example, the lower bound of the CI of the OR crossed 1 at hypoperfusion volumes of ≈ 50 mL for sICH and at ≈ 20 mL for mortality (Figure 3A and 3C). Moreover, for ischemic core volume, the thresholds above which harm was observed may also depend on imaging modality. In the present study, magnetic resonance imaging-based imaging was independently associated with larger estimated core volumes, and recent work on DMVOs has shown that, when CT perfusion is used, the conventional relative cerebral blood flow $<30\%$ threshold may underestimate infarct core.³⁰ Defining clinically meaningful perfusion thresholds for EVT decision-making in iPCAO will require dedicated methodological studies and prospective validation and was beyond the scope of the present study.

This study has limitations. Its retrospective observational design, with treatment decisions based on the choice of the treating physicians, introduces the potential for confounding by indication and baseline imbalances between treatment groups, which may not be fully addressed by statistical adjustment, and unmeasured confounding remains. Selection bias is also likely, as perfusion imaging was not systematically performed across centers and patients. Although baseline characteristics between patients who did and did not undergo perfusion imaging were broadly similar (Table S1), the study population may not be fully representative of all iPCAO cases, and these findings should be interpreted accordingly. In addition, although the overall proportion of missing data was low, complete-case analysis may introduce bias if data are not missing at random, and residual bias related to missingness cannot be excluded. A further limitation is that the cutoff values used to define hypoperfusion and core volume are derived from LVO populations, and growing evidence suggests that these thresholds may not be directly generalizable to other types of stroke.³⁰ Currently, no specific threshold has been established for iPCAO, and the values used in the present study reflect those

applied in routine clinical practice across most centers. Evaluating alternative thresholds was beyond the scope of this work, but further studies are needed to define and validate perfusion thresholds beyond LVO and specific to iPCAO. Additional limitations include unblinded outcome assessment, use of different perfusion software platforms, and mixed CT-based and magnetic resonance imaging-based perfusion imaging although the majority of patients were analyzed using RAPID, and both modalities provide comparable quantitative measures.^{35,36}

In conclusion, in this large international cohort of patients with iPCAO, favorable perfusion characteristics were associated with better clinical outcomes, providing prognostic information independent of EVT use and other baseline characteristics. However, perfusion imaging did not identify any subgroup in which EVT was associated with improved outcomes compared with MM. Instead, perfusion imaging may be more informative for identifying patients at increased risk of harm from EVT than for selecting those likely to benefit, underscoring the need for careful patient selection, and further prospective evaluation of treatment strategies in this population.

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Supplemental Material

Tables S1–S8

Figures S1–S9

STROBE Statement

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